



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - I - Apr - 2026

II Semester

22GE003 & BASICS OF ELECTRICAL ENGINEERING

1	(i)	During a classroom experiment, two plastic rods are rubbed with wool and then brought close to each other. Instead of attracting, the rods move away from each other when placed nearby. Relate this observation to the property of electric charge. (3 Marks - [U/C, 2])
2	(ii)	While studying electrostatics, students observe that some materials become positively charged while others become negatively charged after friction. Identify the basic classification of electric charges. (2 Marks - [U/C, 1])
3	(i)	In an electrostatic demonstration, a teacher charges a metal sphere and then brings it into contact with another neutral sphere. After contact, the charge spreads between the two spheres. When students measure the charges afterward, they find that the combined charge of the two spheres equals the original charge that was present on the first sphere. Even though the charge distribution has changed, the total charge remains unchanged. Establish the law that explains why the total charge remains constant in this interaction. (3 Marks - [U/C, 2])
4	(ii)	In a classroom demonstration, a teacher shows that when several charged bodies interact and exchange electrons, the charges on individual objects may change. However, when all the charges are added together after the interaction, the total value remains exactly the same as before the interaction occurred. This observation helps students understand a fundamental rule governing electric charge in nature. Determine the physical law illustrated in this situation. (2 Marks - [U/C, 1])
5	(i)	In a school physics laboratory, students perform an experiment using a plastic comb and small pieces of paper placed on a table. The comb is first rubbed against dry hair and then brought close to the paper pieces without touching them. The students observe that the paper pieces immediately jump toward the comb and stick to its surface. Even after some pieces fall back to the table, the comb continues to attract other nearby paper fragments. The teacher explains that this behavior occurs because the comb acquires electric charge through friction and produces a force that acts on nearby objects. Establish the role of electric charge in producing forces between objects. (3 Marks - [U/C, 2])
6	(ii)	During a classroom demonstration on electrostatics, a teacher rubs a plastic scale with a dry cloth and then slowly brings it close to tiny bits of thermocol and paper placed on a desk. Students observe that even without touching the pieces, the scale attracts them from a small distance and the particles move toward it. Some pieces briefly stick to the scale before falling off and being attracted again. This repeated interaction helps students understand that a charged body can influence nearby objects through electrostatic forces. Determine the physical property responsible for this electrostatic interaction. (2 Marks - [U/C, 1])

7	(i)	Two metal plates initially carry different amounts of electric charge. When they are connected with a conducting wire, charge flows between them until equilibrium is reached. Relate this observation to the conservation of electric charge. (3 Marks - [U/C, 2])
8	(ii)	During a laboratory experiment, students observe that when electrons move from one conductor to another, the charge on each conductor changes but the overall charge of the system remains unchanged. Identify the law governing this behavior. (2 Marks - [U/C, 1])
9	(i)	During a laboratory investigation of electric charge, scientists observe that whenever an object becomes charged, the charge always appears in fixed discrete amounts rather than any arbitrary value. Even when very small charges are measured, they always appear as multiples of a fundamental unit. This observation leads researchers to study the concept that electric charge exists in discrete packets rather than continuous values. Assess the concept of quantization of electric charge and its significance in electrostatics. (5 Marks - [Ap/P, 2])
10	(i)	In an electronics experiment, a long metallic wire is connected to a battery and a small bulb. When the circuit is completed, the bulb immediately glows, indicating that electric current is flowing through the wire. Scientists explain that inside the wire, millions of electrons move slowly under the influence of the electric field produced by the battery. This gradual movement of electrons in a preferred direction is responsible for current flow. Analyze the role of electron drift in the conduction of electric current in metallic conductors. (5 Marks - [Ap/P, 2])
11	(i)	In an electrostatics experiment, students charge several small metal spheres by rubbing them with different materials. After charging, they bring the spheres into contact with each other and measure the resulting charge using an electrometer. They observe that the final charge on the combined system equals the sum of the individual charges originally present on each sphere. This observation leads them to analyze the principle that electric charges combine algebraically when multiple charged bodies interact. Examine the principle of additivity of electric charges and explain how total charge is determined when several charged bodies interact. (5 Marks - [Ap/P, 2])
12	(i)	During the design of electrical wiring systems in buildings, engineers select materials such as copper and aluminum for conducting electricity. These materials allow electric charges to move easily through them when a voltage is applied. Engineers study the internal structure of these materials to understand why electrons can move freely and how this property allows the conductor to carry electric current efficiently. Assess the behavior of electric charges in conductors and explain why conductors allow the flow of electric current. (5 Marks - [Ap/P, 2])
13	(i)	In a physics laboratory, students perform an experiment using two metal spheres placed on insulating stands. One sphere is positively charged while the other is neutral. When the two spheres are connected with a conducting wire, charge flows between them and eventually distributes evenly. After disconnecting the wire, the students measure the charges on both spheres and notice that although the distribution of charge has changed, the total amount of charge present in the system remains the same as before the interaction. This observation leads them to examine a fundamental law governing electric charge in physical systems. Examine the principle of conservation of electric charge and explain how it applies to the redistribution of charge in a system. (5 Marks - [Ap/P, 2])

14	(i)	In meteorology research centers, scientists investigate the electrical nature of thunderclouds. They observe that ice crystals and water droplets move at different speeds within clouds due to air currents. These movements lead to repeated collisions between particles, causing charge transfer and separation within the cloud. As the cloud grows larger, the accumulation of charges increases the electric potential difference between the cloud and nearby objects such as tall buildings or trees. Analyze how particle collisions inside clouds contribute to the formation of atmospheric electric charges. (5 Marks - [Ap/P, 2])
15	(i)	In particle physics experiments, scientists observe interactions between electrons and protons inside controlled environments. During collisions or interactions, particles may exchange charge or produce other particles, but careful measurements show that the total electric charge before and after the interaction always remains constant. This experimental evidence supports a fundamental conservation principle in physics. Analyze how particle interactions demonstrate the conservation of electric charge. (5 Marks - [Ap/P, 2])
16	(i)	Atmospheric scientists studying thunderstorms use weather balloons and sensors to monitor electrical activity inside clouds. Their observations show that powerful air currents move charged particles upward and downward within the cloud. As these particles collide and separate, large quantities of electric charge accumulate. When the electric field becomes sufficiently strong, a rapid discharge occurs either within the cloud or between the cloud and the Earth's surface. Assess the process by which charge separation in clouds leads to lightning formation. (5 Marks - [Ap/P, 2])
17	(i)	A teaching laboratory places three charges around a measurement point forming a triangular arrangement. Students notice that when one charge is removed, the measured electric field direction changes noticeably even though the remaining two charges remain unchanged. This observation leads them to analyze how individual electric field contributions affect the resultant field. Assess how the principle of superposition explains the change in resultant electric field when one of the charges is removed. (3 Marks - [U/P, 2])
18	(ii)	A charged particle detector is placed near an isolated positive charge. Engineers observe that whenever a small positive test charge is released near the detector it always moves away from the source charge rather than toward it. Infer the cause of this motion using the concept of electric field direction around a point charge. (2 Marks - [U/C, 1])
19	(i)	A square conducting plate carries a total charge of 8 microC distributed uniformly across its surface. The area of the plate is 2 square meters. Students are asked to determine the surface charge density of the plate in order to analyze the electric field generated by the charged surface. Evaluate the surface charge density using the relation $\sigma = Q / A$. (3 Marks - [Ap/P, 2])
20	(ii)	During a classroom experiment a long conducting rod is uniformly charged along its entire length. Students are asked to express the distribution of charge along the rod using a suitable physical quantity. Infer why linear charge density is used to describe this distribution. (2 Marks - [U/C, 1])

21	(i)	During an electrostatic experiment, a charged particle produces an electric field of 12000 N/C at a distance of 1 m from the charge. When the observation point is moved to 2 m away from the same charge, researchers observe that the field intensity decreases significantly. Students are asked to interpret this change based on the mathematical relationship governing electric fields. Determine how the electric field changes with distance and evaluate the new field value at 2 m. (3 Marks - [U/P, 2])
22	(ii)	In a microelectronics laboratory two equal positive charges are placed very close to each other while engineers attempt to find a point between them where the electric field becomes zero. When a third nearby charge is introduced the field at that point no longer cancels. Determine why the electric field cancellation fails. (2 Marks - [U/C, 1])
23	(i)	A simulation program visualizes the electric field produced by a long uniformly charged wire. Students observe that the electric field around the wire forms circular symmetry around the wire and its magnitude decreases as the distance from the wire increases. Researchers explain that the electric field at a point results from the combined contribution of all small charge elements along the wire. Assess why continuous charge distribution requires considering contributions from many small charge elements instead of treating the wire as a single point charge. (3 Marks - [U/P, 2])
24	(ii)	Researchers studying electrostatic systems observe that many real objects such as wires plates and insulating materials contain charges spread over their geometry instead of concentrated at one point. Scientists therefore model these systems using continuous charge distributions. Determine why continuous charge distribution models are important for analyzing electric fields in real systems. (2 Marks - [U/C, 1])
25	(i)	A physics research laboratory is studying electric field visualization using charged metallic spheres. During an experiment, a positively charged sphere is placed in the center of an insulating platform and fine electric field mapping sensors are arranged around it. The sensors record that the electric field appears to spread outward uniformly from the surface of the sphere. When the same experiment is repeated with a negatively charged sphere, the sensors show that the field pattern appears reversed in direction but maintains a similar symmetrical distribution. Researchers use this observation to represent electric fields using field line diagrams. Analyze how the direction and arrangement of electric field lines represent the nature of positive and negative charges and infer why these lines are useful in visualizing electric field behavior. (5 Marks - [Ap/P, 2])
26	(i)	In an electrostatic experiment a very long charged wire carries a uniform linear charge density of $\lambda = 5 \times 10^{-6} \text{ C/m}$. Researchers want to determine the electric field at a point located 0.5 m from the wire. The electric field due to an infinite line charge is given by $E = \lambda / (2 \times \pi \times \epsilon_0 \times r)$ where $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$. Determine the magnitude of the electric field at the observation point. (5 Marks - [Ap/P, 2])

27	(i)	An electrostatic simulation system is used to study the interaction between two identical positive charges placed close to each other. When the electric field pattern is generated, students observe that the electric field lines originating from each charge curve away from the region between them rather than crossing directly toward the other charge. The space between the charges shows a region where the lines appear to repel each other. Evaluate why electric field lines behave in this manner and determine what this pattern indicates about the interaction between the two positive charges. (5 Marks - [Ap/P, 2])
28	(i)	An electrostatics laboratory studies the electric field produced by very long charged conductors. In one experiment, researchers place a very long straight wire carrying uniform positive charge and measure the electric field at different points around it using sensors. They observe that the electric field magnitude depends only on the radial distance from the wire and remains the same at all points located at the same distance around it. Scientists explain that the symmetry of the charge distribution plays an important role in determining the electric field pattern. Analyze how the symmetry of an infinitely long charged wire influences the electric field distribution and evaluate why the electric field depends only on the perpendicular distance from the wire. (5 Marks - [Ap/P, 2])
29	(i)	In an electrostatic experiment two point charges of equal magnitude but opposite sign are placed 4 cm apart forming an electric dipole. Each charge has a magnitude of 3 microC. Researchers want to calculate the electric dipole moment of this system in order to understand the strength of the dipole. Determine the electric dipole moment using the relation $p = q \times d$ (5 Marks - [Ap/P, 2])
30	(i)	Scientists analyze the electric field produced by a hollow spherical shell carrying uniform positive charge on its surface. They place electric field probes at three regions: inside the hollow cavity, on the surface of the shell, and at points outside the shell. Measurements show that the electric field is zero inside the cavity, reaches a certain value at the surface, and decreases as the distance from the sphere increases. Analyze why the electric field behaves differently in these three regions and infer how the spherical symmetry influences the field distribution. (5 Marks - [Ap/P, 2])
31	(i)	A nanotechnology laboratory is studying the behavior of molecular systems that exhibit electric dipole characteristics. In one experiment, scientists place two equal but opposite charges separated by a small fixed distance on an insulating platform to simulate a dipole. Sensors placed around the setup detect that the electric field distribution in the surrounding space differs significantly from that produced by a single charge. The field pattern shows strong directional variation, and the effect becomes more pronounced when the separation distance between the charges increases while maintaining the same charge magnitude. Analyze how the presence of two equal and opposite charges forms an electric dipole and evaluate how the separation distance influences the dipole's electric field behavior. (5 Marks - [Ap/P, 2])

32	(i)	A uniformly charged spherical shell carries a total charge of $Q = 6 \times 10^{-6} \text{ C}$ and has a radius of $R = 0.5 \text{ m}$. Researchers want to calculate the electric field at a point located $r = 1 \text{ m}$ from the center of the shell. The electric field outside a spherical shell is given by $E = (1 / (4 \times \pi \times \epsilon_0)) \times (Q / r^2)$ where $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$. Determine the electric field magnitude at the observation point. (5 Marks - [Ap/P, 2])
33	(i)	An engineer testing an electromagnet observes that increasing current in the coil significantly strengthens the magnetic field produced by the device. The electromagnet is connected to a DC supply and the current flowing through the coil increases from 2 A to 6 A while the coil geometry remains unchanged. The engineer investigates the relationship between electric current and magnetic field generation around the conductor. Formula Hint: $B \propto I$ Analyze why increasing current increases magnetic field strength in the electromagnet. Infer how current variation influences electromagnetic device performance. (3 Marks - [An/C, 2])
34	(ii)	A designer working on electromagnetic lifting equipment studies the magnetic field produced by current-carrying coils. The designer notes that stronger magnetic fields improve lifting capacity and considers increasing the coil current in the electromagnet. Determine how current variation affects magnetic field generation around the coil. Infer how this principle supports the design of electromagnetic lifting systems. (2 Marks - [An/C, 2])
35	(i)	An atomic model shows an electron revolving in a circular orbit around the nucleus. The direction of the resulting magnetic dipole moment depends on the direction of the electron's motion. Scientists use a rule similar to the right-hand rule to determine the direction of the magnetic dipole moment generated by the moving charge. Determine how the direction of the magnetic dipole moment is related to the electron's orbital motion. (3 Marks - [An/C, 2])
36	(ii)	A circular electron orbit has radius $4 \times 10^{-11} \text{ m}$ and carries an effective current of $2 \times 10^{-3} \text{ A}$. Formula Hint: Magnetic Dipole Moment $\mu = I \times A$ Area $A = \pi r^2$ Determine the magnetic dipole moment of the orbiting electron. (2 Marks - [Ap/P, 2])
37	(i)	A student observes magnetic field lines around a bar magnet placed near iron filings on a laboratory table. The filings arrange themselves along curved patterns extending from the north pole to the south pole of the magnet. The student notes that these patterns visually represent the direction and strength of the magnetic field around the magnet. Determine how magnetic field lines represent the direction of magnetic forces around a magnet. Evaluate how the density of field lines indicates the strength of the magnetic field. (3 Marks - [An/C, 2])

38	(ii)	A magnetic core inside a transformer experiences a magnetic flux of 0.02 Weber passing through a cross-sectional area of 0.01 m². Engineers need to determine the magnetic flux density inside the core to evaluate the performance of the transformer core material. Formula Hint: $B = \Phi / A$ Determine the magnetic flux density in the transformer core. (2 Marks - [Ap/C, 2])
39	(i)	A student observes that atoms containing unpaired electrons often exhibit magnetic behavior, while atoms with paired electrons usually show weak or no magnetism. This observation relates to the way magnetic dipole moments of electrons combine within an atom. Determine why paired electrons cancel magnetic effects while unpaired electrons contribute to atomic magnetism. (3 Marks - [An/C, 2])
40	(ii)	In atomic magnetism, the magnetic dipole moment is directly proportional to the current and the area of the orbit. Formula Hint: $\mu = I \times A$ Determine how increasing the radius of the electron orbit affects the magnetic dipole moment. (2 Marks - [An/C, 2])
41	(i)	A laboratory experiment measures the magnetic field produced around a long straight copper conductor carrying a steady current of 12 A. The measurement probe is placed at a distance of 0.03 m from the wire. Engineers want to determine the magnetic field intensity at this point using the theoretical formula derived from Ampere's Circuital Law. Formula Hint: $B = (\mu_0 \times I) / (2 \times \pi \times r)$ $\mu_0 = 4 \times \pi \times 10^{-7}$ Determine the magnetic field intensity at the specified point and analyze how reducing the distance from the conductor would affect the magnetic field strength. (5 Marks - [Ap/P, 2])
42	(i)	A small bar magnet used in a laboratory experiment behaves like a magnetic dipole. The magnetic moment of the magnet is 0.5 A·m², and a measurement probe is placed 0.10 m away from the center of the magnet along its axial line. Engineers studying magnetic field distribution want to determine the magnetic field intensity at this point. Understanding the axial magnetic field helps in designing magnetic sensors and analyzing magnetic material behavior. Formula Hint: Magnetic field on axial line of dipole: $B_{\text{axial}} = (\mu_0 / 4\pi) \times (2M / r^3)$ $\mu_0 / 4\pi = 10^{-7}$ Determine the magnetic field intensity along the axial line and analyze how increasing the distance affects the field strength. (5 Marks - [Ap/P, 2])

43	(i)	An engineering student writes a program to calculate the magnetic field around a current-carrying wire but forgets to complete some statements. The program should calculate the magnetic field intensity at a point located a certain distance from the wire using Ampere's law. The program structure is partially written as shown below. Program Snippet: Input: Current I, Distance r Constant $\mu_0 = 4 \times \pi \times 10^{-7}$ Step1: _____ Step2: $B = (\mu_0 \times I) / (2 \times \pi \times r)$ Step3: _____ Formula Hint: $B = (\mu_0 \times I) / (2 \times \pi \times r)$ Determine the missing steps in the program so that the system correctly calculates and displays the magnetic field. (5 Marks - [Ap/P, 2])
44	(i)	A magnetic dipole with magnetic moment $0.4 \text{ A}\cdot\text{m}^2$ is placed in a laboratory setup for studying magnetic field distribution. A magnetic field sensor is placed at a point 0.08 m from the center of the magnet along the equatorial line, which is perpendicular to the magnet's axis. Engineers need to determine the magnetic field intensity at this location to compare axial and equatorial magnetic field strengths. Formula Hint: Magnetic field on equatorial line: $B_{\text{equatorial}} = (\mu_0 / 4\pi) \times (M / r^3)$ $\mu_0 / 4\pi = 10^{-7}$ Determine the magnetic field intensity along the equatorial line and interpret how it compares with the axial field. (5 Marks - [Ap/P, 2])
45	(i)	Two solenoids operate with the same current 2 A. Solenoid X has 1500 turns/m, while Solenoid Y has 2500 turns/m. Formula Hint: $B = \mu_0 \times n \times I$ Determine the magnetic fields of both solenoids and identify which produces the stronger magnetic field. (5 Marks - [Ap/P, 2])
46	(i)	A coil with 120 turns experiences a decrease in magnetic flux from 0.03 Weber to 0.01 Weber in 0.02 seconds due to movement of a magnet away from the coil. Formula Hint: $E = -N \times (\Delta\Phi / \Delta t)$ Determine the induced EMF in the coil. (5 Marks - [Ap/P, 2])
47	(i)	A solenoid with 1000 turns is wound uniformly over a length of 0.50 m. The coil carries 1.8 A current. Engineers need to calculate the magnetic field generated inside the solenoid. Formula Hint: $B = \mu_0 \times n \times I$ $n = N / L$ $\mu_0 = 4\pi \times 10^{-7}$ Determine the magnetic field strength inside the solenoid. (5 Marks - [Ap/P, 2])

48	(i)	Two coils experience identical magnetic flux change rate of 0.02 Wb/s. Coil X has 150 turns, while Coil Y has 300 turns. Formula Hint: $E = - N \times (d\Phi / dt)$ Determine the induced EMF in both coils. (5 Marks - [Ap/P, 2])
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